

Noether Sheet 8

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These were my solutions to the Noether Mentoring Scheme Sheet 8. Problem statements are abridged.

Problem 1

Suppose that

$$x^2 - 3ax + b = 0 \text{ has solutions } x = c \text{ and } x = d, \text{ and}$$

$$x^2 + cx + d = 0 \text{ has solution } x = a \text{ and } x = b.$$

All of a, b, c and d are non-zero. What are the possible values of a, b, c and d ?

Solution

Vieta's formulae give us the following relations

$$3a = c + d, \quad b = cd$$

$$-c = a + b, \quad d = ab$$

Using the two equations on the right we find that

$$b = c(ab) \iff ac = 1 \text{ (Since } b \neq 0 \text{)}$$

Now observe

$$\begin{aligned} 3a &= c + d \\ &= c + (ab) \\ &= c + a(-a - c) \\ &= c - ac - a^2 \\ &= c - (1) - a^2 \\ 3a^2 &= (ac) - a - a^3 \\ &= 1 - a - a^3 \end{aligned}$$

So a satisfies $a^3 + 3a^2 + a - 1 = 0$. By inspection $a = -1$ is a root. Now we can use polynomial long division to find that

$$(a + 1)(a^2 + 2a - 1) = 0$$

so $a \in \{-1, -1 + \sqrt{2}, -1 - \sqrt{2}\}$.

Case 1: $a = -1$. Then $c = -1$. Also $d = 3(-1) - (-1) = -2$. Finally $b = -(-1) - (-1) = 2$. Let's check that this works,

$$3(-1) = (-1) + (-2), \quad 2 = (-1)(-2)$$

$$-(-1) = (-1) + (2), \quad -2 = (-1)(2)$$

so $(a, b, c, d) = (-1, 2, -1, -2)$ is a solution.

Case 2: $a = -1 + \sqrt{2}$. Then $c = 1/(-1 + \sqrt{2}) = 1 + \sqrt{2}$. Also $d = 3(-1 + \sqrt{2}) - (1 + \sqrt{2}) = -4 + 2\sqrt{2}$. Finally $b = -(-1 + \sqrt{2}) - (1 + \sqrt{2}) = -2\sqrt{2}$. Let's check that it works.

$$3(-1 + \sqrt{2}) = (1 + \sqrt{2}) + (-4 + 2\sqrt{2}), \quad -2\sqrt{2} = (1 + \sqrt{2})(-4 + 2\sqrt{2})$$

$$-(1 + \sqrt{2}) = (-1 + \sqrt{2}) + (-2\sqrt{2}), \quad -4 + 2\sqrt{2} = (-1 + \sqrt{2})(-2\sqrt{2})$$

so $(a, b, c, d) = (-1 + \sqrt{2}, -2\sqrt{2}, 1 + \sqrt{2}, -4 + 2\sqrt{2})$ is a solution.

Case 3: $a = -1 - \sqrt{2}$. Then $c = 1/(-1 - \sqrt{2}) = 1 - \sqrt{2}$. Also $d = 3(-1 - \sqrt{2}) - (1 - \sqrt{2}) = -4 - 2\sqrt{2}$. Finally $b = -(-1 - \sqrt{2}) - (1 - \sqrt{2}) = 2\sqrt{2}$. Let's check that

$$3(-1 - \sqrt{2}) = (1 - \sqrt{2}) + (-4 - 2\sqrt{2}), \quad 2\sqrt{2} = (1 - \sqrt{2})(-4 - 2\sqrt{2})$$

$$-(1 - \sqrt{2}) = (-1 - \sqrt{2}) + (2\sqrt{2}), \quad -4 - 2\sqrt{2} = (-1 - \sqrt{2})(2\sqrt{2})$$

so $(a, b, c, d) = (-1 - \sqrt{2}, 2\sqrt{2}, 1 - \sqrt{2}, -4 - 2\sqrt{2})$ is a solution. ■

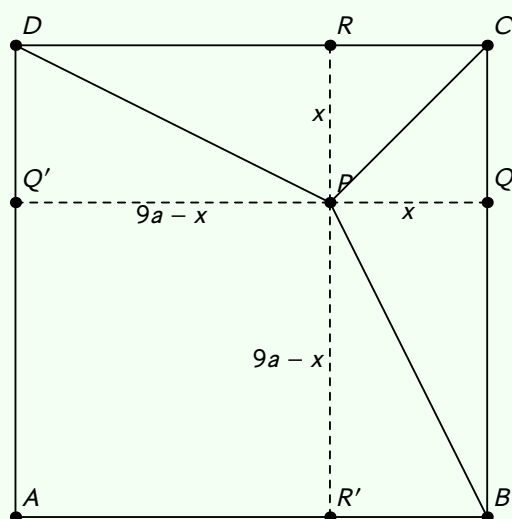
Problem 2

A square field has vertices at $(0,0)$, $(9a,0)$, $(0,9a)$ and $(9a,9a)$. A point P is selected in the field, and three walls are drawn to the edge of the field, dividing it into two triangles and a quadrilateral. All three regions have the same area.

How many possible locations of P are there?

Solution

(Sketch of solution.) The answer is that there are four locations for P . The three walls must each be to vertices of the square because otherwise we won't have a quadrilateral but some other n -gon ($n > 4$). Consider the diagram where $A = (0,0)$ is the bottom left corner of the square.



Note that $PQ = RP$ because the two triangles $\triangle DCP$ and $\triangle BPC$ have the same base length and the same area, so their perpendicular heights must also be the same. We call this length x .

The area of each triangle is $\frac{1}{2}9ax$. The area of the quadrilateral is $\frac{1}{2}9a(9a - x) + \frac{1}{2}9a(9a - x)$. Since all three regions have equal areas,

$$\frac{1}{2}9ax = 9a(9a - x)$$

$$x = 2(9a - x)$$

$$3x = 18a$$

$$x = 6a$$

So that means that $P = (3a, 3a)$. Now you can rotate and reflect this diagram accordingly to find that for other configurations we could have

$$P = (3a, 3a), (6a, 6a), (6a, 3a), (3a, 6a). \blacksquare$$

Problem 3

Let S be the set of numbers formed by ordering five different digits in each possible order. Show that there must be two elements s and t of S such that either $s + t$ or $s - t$ is divisible by 233.

Solution

For clarity I am assuming that we are disregarding any leading zeros. For example if the five different digits happen to be 0, 3, 4, 6, 9 then I'll assume that the number 03469 is to be interpreted as 3469 and this number appears in S .

Our strategy is that we are going to use the Pigeonhole principle. All residues modulo 233 are a part of the set $\{0, \pm 1, \pm 2, \dots, \pm 116\}$, so if we have a set of ≥ 118 numbers there must be two s, t such that $s \equiv \pm t \pmod{233}$.

There are $5!$ ways to order the digits and so $|S| = 5! = 120$. Now note that every number modulo 233 is a part of the set $\{0, \pm 1, \pm 2, \dots, \pm 116\}$. Ignoring the $\pm$ signs there are 117 numbers in this set. So if we have a group of $120 > 117$ numbers then, by the Pigeonhole principle there must be two numbers s and t such that $s \equiv \pm t \pmod{233}$. But that means that one of $s - t$ or $s + t$ is divisible by 233, as desired. ■

Comments: Though it may look like number theory upon a first glance, once you read the problem statement it feels too strange to be number theory. The number 233 has no remarkable properties, so it is natural to think about a possible combinatorial approach given that the problem talks about ordering digits. Finally once you spot the pigeonhole principle the problem collapses.

Problem 4

Let $F = \{1, 2, 3, 5, 8, \dots\}$ denote the set of Fibonacci numbers. Then let $F + F$ denote the set of all numbers that are the sum of two fibonacci numbers.

Show that there are infinitely many arithmetic progressions of length 6 in $F + F$.

Thoughts

The following construction almost works

$$\begin{aligned} F_{n-2} + F_{n-1} &= F_n \\ F_1 + F_n &= F_n + 1 \\ F_2 + F_n &= F_n + 2 \\ F_3 + F_n &= F_n + 3 \\ ? + ? &= F_n + 4 \\ F_4 + F_n &= F_n + 5 \end{aligned}$$

so it is natural to ask whether $F_n + 4$ can be expressed as a sum of two fibonacci numbers, if we can find some condition where this is true we will have solved the problem. After experimenting it seems that apart from some small cases usually it isn't possible to write $F_n + 4$ as the sum of two fibonacci numbers. Actually nevermind consider the following thought.

One class of numbers that occurs in $F + F$ is numbers of the form $F_n + F_n = 2F_n$. So if $\frac{F_n+4}{2}$ happens to be some fibonacci number F_j this would be enough to solve the problem. Once again after experimenting this doesn't seem to lead anywhere, but it could be the case that such numbers F_j are just really sparse. Actually I think that this approach really is unrepairable since it is known that the fibonacci numbers are roughly exponential ($F_{n+1}/F_n \rightarrow \varphi \approx 1.618\dots$ as n gets large), I think I can show that for large enough n It's not possible to write $F_n + 4$ as the sum of two fibonacci numbers.

After this point I thought, perhaps there is some nice fibonacci identity that will give me some sort of construction on how to find these arithmetic progressions of fibonacci numbers. But after some thought I found nothing useful to the task at hand, so I resorted more experimentation. This time I thought about this: "What numbers CAN'T be expressed as the sum of two fibonacci numbers?"

$2 = 1 + 1$	$17 = x$	$35 = 34 + 1$	$54 = x$
$3 = 2 + 1$	$18 = 13 + 5$	$36 = 34 + 2$	$55 = 21 + 34$
$4 = 3 + 1$	$19 = x$	$37 = 21 + 3$	$56 = 55 + 1$
$5 = 3 + 2$	$20 = x$	$38 = x$	$57 = 55 + 2$
$6 = 5 + 1$	$21 = 8 + 13$	$39 = 34 + 5$	$58 = 55 + 3$
$7 = 5 + 2$	$22 = 21 + 1$	$40 = x$	$59 = x$
$8 = 5 + 3$	$23 = 21 + 2$	$41 = x$	$60 = 55 + 5$
$9 = 8 + 1$	$24 = 21 + 3$	$42 = 34 + 8$	$61 = x$
$10 = 8 + 2$	$25 = x$	$43 = x$	$62 = x$
$11 = 8 + 3$	$26 = 21 + 5$	$44 = x$	$63 = 55 + 8$
$12 = x$	$27 = x$	$45 = x$	$64 = x$
$13 = 8 + 5$	$28 = x$	$46 = x$	$65 = x$
$14 = 13 + 1$	$29 = 21 + 8$	$47 = 34 + 13$	$66 = x$
$15 = 13 + 2$	$30 = x$	$48 = x$	$67 = x$
$16 = 13 + 3$	$31 = x$	$49 = x$	$68 = 55 + 13$
	$32 = x$	$50 = x$	$69 = x$
	$33 = x$	$51 = x$	
	$34 = 13 + 21$	$52 = x$	
		$53 = x$	

I noticed the following. If we order the elements of $F + F$ then it "usually" looks like

$$F_n, F_n + F_1, F_n + F_2, \dots, F_n + F_{n-1}.$$

I also see that, apart from some small values, most numbers cannot be expressed as the sum of two fibonacci numbers.

From this point on I changed my mindset about this problem. There is a large class of mathematics problems that can be solved by the technique of "spot this pattern and win", but looking at the data of this problem, it seems that there are no real patterns. The number 6 in the problem statement seems rather arbitrary, and it could very well be that case that the set of fibonacci numbers is a red herring. It leads me to think that the following statement might be true,

Let S be an infinite set of positive integers. Let S' be a set such that if $a, b \in S$ (not necessarily distinct), then $a + b \in S'$. Then there are infinitely many arithmetic progressions of length k in S' .

In particular for our problem we want to show the statement for $k = 6$. This is a hard problem. Let's instead try and prove/disprove the smallest case, that is $k = 3$. In fact when $k = 3$ the problem becomes easy (relatively speaking). Here are the details.

Suppose the numbers a and $a + \delta$ are both in S . Also the numbers

$$(a) + (a) = 2a, \quad (a) + (a + \delta) = 2a + \delta, \quad (a + \delta) + (a + \delta) = 2a + 2\delta$$

are all in S' . This is an arithmetic progression of length 3, done!

Now what about $k = 4$? I'm pretty sure that the statement is false for $k \geq 4$ after some experimentation. Back to the drawing board then.

I remember from Noether Sheet 6, the following identity $3F_n = F_{n+2} + F_{n-2}$. So both $2F_n, 3F_n \in F + F$. It is natural to hope that maybe $4F_n, 5F_n$ etc appear in $F + F$, but again after some experimentation I don't

believe these multiples of F_n occur "regularly". After a while of messing around I came across the fact that

$$F_{n-3}, F_{n-1}, F_n, F_n + F_{n-2}$$

are all in $F + F$ for large enough n and this sequence is an AP with common difference F_{n-2} . If we can show that it's possible to get "lucky" infinitely often, that is find solutions to

$$F_n + 2F_{n-2} = \text{Sum of two fibonacci numbers}$$

$$F_n + 3F_{n-2} = \text{Sum of two fibonacci numbers}$$

then we would be done. For the first equation I naturally considered the following table

F_n	$F_n + 2F_{n-2}$	Sum of two ?
3	$3 + 2(1) = 5$	$2 + 3$
5	$5 + 2(2) = 9$	$8 + 1$
8	$8 + 2(3) = 14$	$13 + 1$
13	$13 + 2(5) = 23$	$21 + 2$
21	$21 + 2(8) = 37$	$34 + 3$
34	$34 + 2(13) = 60$	$55 + 5$
55	$55 + 2(21) = 97$	$89 + 8$
89	$89 + 2(34) = 157$	$144 + 13$
144	$144 + 2(55) = 254$	$233 + 21$

I continued this until $F_n = 1597$ and every time it was possible to express $X = F_n + 2F_{n-2}$ as the sum of two fibonacci numbers, so it's very likely that there is an identity to always express X as the sum of two fibonacci numbers. After looking at the table we conjecture that $F_n + 2F_{n-2} = F_{n+1} + F_{n-4}$, and it's quite easy to show that this result is true. The last part is just to express $F_n + 3F_{n-2}$ as the sum of two fibonacci numbers. Unfortunately if we try to make the same table for $F_n + 3F_{n-2}$ it seems that usually numbers of this form can't be expressed as the sum of two fibonacci numbers.

Finally after the hint to look at the $a + \delta$ logic but just replace this with fibonacci numbers and their identities I found a working construction. That is

$$\underbrace{F_{n-4}}_{=2F_{n-2}-F_{n-1}}, F_{n-2} + F_{n-2}, F_n + F_{n-2}, F_n + F_n, F_n + F_{n+1}, F_{n+1} + F_{n+1}$$

which is an AP of length 6, with common difference F_{n-1} .

Solution

We show the desired result by a construction.

Claim 1: Let $n \geq 4$ be an integer. Then $F_{n-4} = 2F_{n-2} - F_{n-1}$.

Proof: Observe that the RHS is

$$\begin{aligned} 2F_{n-2} - F_{n-1} &= F_{n-2} + (F_{n-2} - F_{n-1}) \\ &= F_{n-2} + (F_{n-2} - [F_{n-2} + F_{n-3}]) \\ &= F_{n-2} - F_{n-3} \\ &= F_{n-4} \end{aligned}$$

as desired. $\square$

Let $n \geq 6$ be an integer. Then notice that we can write $F_{n-4} = F_{n-5} + F_{n-6}$. Now consider the following sequence.

$$F_{n-5} + F_{n-6}, \quad F_{n-2} + F_{n-2}, \quad F_n + F_{n-2}, \quad F_n + F_n, \quad F_n + F_{n+1}, \quad F_{n+1} + F_{n+1}.$$

Each term is F_{n-1} more than the last and clearly all 6 terms are in $F + F$. Thus there are infinitely many AP's of length 6 in $F + F$. $\blacksquare$

Comments: Although the solution is really short this really isn't an easy problem.

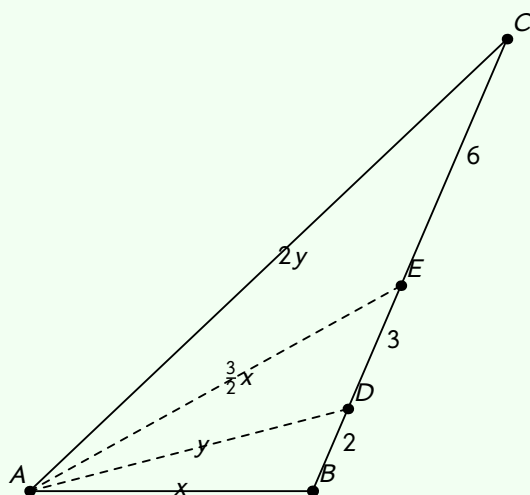
Problem 5

In triangle ABC , the angle $\angle BAC$ is trisected by AD and AE , where D and E lie on BC . Suppose that $BD = 2$, $DE = 3$ and $EC = 6$.

What is the length of AC ?

Solution

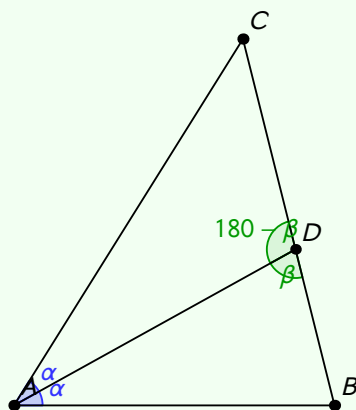
We have the diagram



Denote $x = AB$ and $y = AD$. We are going to justify the other lengths with this theorem.

Angle Bisector Theorem: Let $\triangle ABC$ have a bisector AD , where D lies on BC . Then $DB : DC = AB : AC$.

Proof: Consider,



By sine rule

$$\frac{DB}{\sin \alpha} = \frac{AB}{\sin \beta}, \quad \frac{CD}{\sin \alpha} = \frac{AC}{\sin(180 - \beta)} = \frac{AC}{\sin \beta}$$

We can re-arrange to get

$$\frac{DB}{AB} = \frac{CD}{AC} = \frac{\sin \alpha}{\sin \beta} \Rightarrow \frac{DB}{DC} = \frac{AB}{AC}. \blacksquare$$

Now since AD bisects $\angle BAE$ we deduce that $AE = \frac{3}{2}x$. Similarly we find that $AC = 2y$.

Now by applying Cosine rule on $\triangle ADB$, $\triangle ADE$ and $\triangle AEC$ we have the following equations

$$4 = x^2 + y^2 - 2xy \cos \theta$$

$$9 = \frac{9}{4}x^2 + y^2 - 3xy \cos \theta$$

$$36 = \frac{9}{4}x^2 + 4y^2 - 6xy \cos \theta$$

Now one can use whatever method they like to solve this system in terms of x^2 , y^2 and $xy \cos \theta$, I'll use matrices.

$$\begin{bmatrix} 1 & 1 & -2 \\ 9/4 & 1 & -3 \\ 9/4 & 4 & -6 \end{bmatrix} \begin{bmatrix} x^2 \\ y^2 \\ xy \cos \theta \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 36 \end{bmatrix}$$

$$\begin{bmatrix} x^2 \\ y^2 \\ xy \cos \theta \end{bmatrix} = \begin{bmatrix} 40 \\ 54 \\ 45 \end{bmatrix}$$

which yields $x = 2\sqrt{10}$, $y = 3\sqrt{6}$ and $\cos \theta = \sqrt{15}/4$. Finally the problem asked us for the length of AC which is $2y = 6\sqrt{6}$. $\blacksquare$

Problem 6

If x, y and z are non-negative numbers, prove that at least one of $x - xy$, $y - yz$ and $z - zx$ is at most $\frac{1}{4}$.

Solution

Assume the contrary, that is

$$x(1 - y) > \frac{1}{4}, \quad y(1 - z) > \frac{1}{4}, \quad z(1 - x) > \frac{1}{4}$$

Then multiply all three inequalties together to get

$$x(1 - x)y(1 - y)z(1 - z) > \frac{1}{64}.$$

Define $f(x) = x(1 - x)$. It is routine to check that f has a maxima when $x = \frac{1}{2}$ yielding $f(1/2) = 1/4$. So we deduce that

$$\frac{1}{64} < x(1 - x)y(1 - y)z(1 - z) \leq \frac{1}{64}.$$

but this implies that

$$\frac{1}{64} < \frac{1}{64}$$

a contradiction! ■

Problem 7

An unreliable postman has seven letters to deliver, one to each of seven houses. In how many ways can they do this such that no house receives the correct letter?

Solution

First we remove all creative sparkle from the problem. Let $f(n)$ count the number of permutations σ of $\{1, 2, \dots, n\}$ such that σ has no fixed points. The problem is just asking us to compute $f(7)$. Let $c(n)$ count the number of permutations σ of $\{1, 2, \dots, n\}$ such that σ has no cycles of length $< n$ and no fixed points. It will prove useful to us to find a closed form for $c(n)$.

Claim 1: $c(n) = (n-1)!$.

Proof: Call a permutation σ of $\{1, 2, \dots, n\}$ *nice* if it has no cycles of length $< n$ and no fixed points. Suppose that the permutation σ is nice, notice that the following is true. For any $x \in \{1, 2, \dots, n\}$, the smallest integer k satisfying $\sigma^k(x) = x$ is $k = n$. This is because if $k < n$ then we have found a cycle of length $< n$. This implies that

$$\{\sigma(1), \sigma^2(1), \dots, \sigma^n(1)\} = \{1, 2, \dots, n\} \quad (*)$$

and also the same is true when all the $\sigma^k(1)$'s are replaced with $\sigma^k(x)$ for any $x = 1, 2, \dots, n$. We know that $\sigma^n(1)$ must be $= 1$, but otherwise we can choose the values of $\sigma(1), \sigma^2(1), \dots, \sigma^{n-1}(1)$ freely, they just have to form a permutation of $\{2, 3, \dots, n\}$. There are $(n-1)!$ ways to do so, to finish we should show that this action also uniquely determines the values of $\sigma(2), \sigma(3), \dots, \sigma(n)$.

For any $x = 2, 3, \dots, n$ there exists a unique $k = 1, 2, \dots, (n-1)$ such that $\sigma^k(1) = x$ due to $(*)$. Thus $\sigma(x) = \sigma^{k+1}(1)$ and since $\sigma^2(1), \sigma^3(1), \dots, \sigma^n(1)$ are all distinct (again due to $(*)$) we have indeed uniquely determined $\sigma(2), \sigma(3), \dots, \sigma(n)$. $\square$

Let σ be a permutation of $\{1, 2, \dots, 7\}$ such that σ has no fixed points. Since it has no fixed points any cycles in σ must have length > 1 . It can be checked that there are exactly four ways to organise the number of cycles in σ , they are

$$(7), (2, 2, 3), (2, 5), (3, 4).$$

So that means we could have one cycle of length 7, or two cycles of length 2 with one cycle of length 3, etc... From here we have all we need to compute $f(7)$,

$$\begin{aligned} f(7) &= \binom{7}{7} \cdot c(7) + \binom{7}{2} \binom{5}{2} \binom{3}{3} \cdot c(2) \cdot c(2) \cdot c(3) \\ &\quad + \binom{7}{2} \binom{5}{5} \cdot c(2) \cdot c(5) + \binom{7}{3} \binom{4}{4} \cdot c(3) \cdot c(4) \\ &= 2064. \blacksquare \end{aligned}$$

Addendums (after having read solutions).

After seeing that my solution was wrong I tried the problem again. Call a permutation *nice* if it has no fixed points.

Claim 1: Consider the permutations $\sigma_1, \sigma_2, \dots, \sigma_{n!}$ of $\{1, 2, \dots, n\}$. The $(n+1)!$ permutations of $\{1, 2, \dots, n+1\}$ can be generated via the following algorithm.

1. For each permutation σ_i , ($i = 1, 2, \dots, n!$) do the following,
2. For each ($k = 1, 2, \dots, (n+1)$) create a new permutation, call it $\mu_{i,k}$ such that $\mu_{i,k}(n+1) = k$ and $\mu_{i,k}(k) = n+1$ and for all other ($x = 1, 2, \dots, (n+1)$), $x \neq (n+1)$, $x \neq k$ set $\mu_{i,k}(x) = \sigma_i(x)$.

Informally that is to let $\mu_{i,k}$ swap the position of k and $(n+1)$ and for all the other numbers x let $\mu_{i,k}(x) = \sigma_i(x)$.

Proof: The algorithm loops over $n! \times (n+1)$ times and so creates $(n+1)!$ permutations of length $(n+1)$. Consider the permutations generated $\mu_{1,1}, \mu_{1,2}, \dots, \mu_{n!,(n+1)}$. We must show that the permutations in this list are pairwise distinct. Define $S_i = \{\sigma_i(t) : t = 1, 2, \dots, n\}$ and $S_j = \{\sigma_j(t) : t = 1, 2, \dots, n\}$.

First we show that if $i \neq j$ then $\mu_{i,x} \neq \mu_{j,y}$. Note that σ_i and σ_j are not the same permutation. There must exist (at least) two different numbers $1 \leq a, b \leq n$ such that

$$\sigma_i(a) \neq \sigma_j(a) \text{ and } \sigma_i(b) \neq \sigma_j(b).$$

To see why note that if there was only one number k such that $\sigma_i(k) \neq \sigma_j(k)$, then $|S_i \cap S_j| = n-1$. Since permutations are surjective, both $|S_i| = n$ and $|S_j| = n$. Then it is a well-known fact that

$$\begin{aligned} |S_i \cup S_j| &= |S_i| + |S_j| - |S_i \cap S_j| \\ &= n + n - (n-1) \\ &= n+1 \end{aligned}$$

but $|S_i \cup S_j|$ should be $= n$, contradiction. Now back to the main goal, since $a \neq b$ at least one of a, b is not equal to k . Suppose WLOG that $a \neq k$. Then because of how the algorithm constructs permutations we have that $\mu_{i,x}(a) \neq \mu_{j,y}(a)$. This is because both $\mu_{i,x}(a)$ and $\mu_{j,y}(a)$ will be defined to be $\sigma_i(a), \sigma_j(a)$ respectively, as can be seen in the latter part of step 2 of the algorithm.

Next we show that if $x \neq y$ then $\mu_{i,x} \neq \mu_{j,y}$. This proof is much easier than the last case, just note that $\mu_{i,x}(x) = n+1$ whilst $\mu_{j,y}(y) = n+1$ and since $x \neq y$ they clearly aren't the same permutation.

So we have shown that if $\mu_{i,x} = \mu_{j,y}$ then $i = j$ and $x = y$. This guarantees that the $(n+1)!$ permutations generated are in fact pairwise distinct, as desired. $\square$

Consider some step of the algorithm when the permutation σ_i has > 1 fixed point. We claim that no nice permutations $\mu_{i,k}$ can be generated for this i . Suppose that a, b are fixed points of σ_i permutation. Formally $\sigma_i(a) = a$ and $\sigma_i(b) = b$.

- If k is not equal to either of a, b then $\mu_{i,k}(a) = \sigma_i(a) = a$ and $\mu_{i,k}(b) = \sigma_i(b) = b$, which is due to step 2 of the algorithm, and so $\mu_{i,k}$ isn't nice.
- If k is equal to one of a, b , say WLOG $k = a$, then $\mu_{i,k}(b) = \sigma_i(b) = b$, which is due to step 2 of the algorithm, and so $\mu_{i,k}$ isn't nice.

Suppose σ_i has exactly one fixed point a , Informally we just need to ensure $\mu_{i,k}$ swaps the position of a and $(n+1)$ and then for all the remaining numbers setting $\mu_{i,k}(x) = \sigma_i(x)$ guarantees no fixed points. More formally, in order to get rid of this fixed point in $\mu_{i,k}$ we need $k = a$ and that way we have $\mu_{i,k}(a) = n+1$ and $\mu_{i,k}(n+1) = a$, and for the remaining numbers ($x = 1, 2, \dots, n$), ($x \neq k$), ($x \neq (n+1)$) we have $\mu_{i,k}(x) = \sigma_i(x) \neq x$ since we assumed there was only one fixed point in σ_i . Thus we know that $\mu_{i,k}$ is nice.

Suppose that σ_i has no fixed points. Informally we are going to note that if $\mu_{i,k}$ swaps $(n+1)$ with any other number then the resulting permutation is still nice. More formally if we let $k = 1, 2, \dots, n$ the permutation $\mu_{i,k}$ is nice, because we will get $\mu_{i,k}(k) = n+1$ and $\mu_{i,k}(n+1) = k$ and for the remaining numbers $(x = 1, 2, \dots, n), (x \neq k), (x \neq (n+1))$ we know that $\mu_{i,k}(x) = \sigma_i(x) \neq x$ since σ_i has no fixed points. Thus $\mu_{i,k}$ is nice.

Hence if $f(n)$ is the number of nice permutations of length n then,

$$f(n+1) = \# \left(\begin{array}{c} \text{Permutations of length } n \\ \text{with exactly 1 fixed point.} \end{array} \right) + \underbrace{n \times \# \left(\begin{array}{c} \text{Permutations of length } n \\ \text{with no fixed points.} \end{array} \right)}_{\text{There are } n \text{ ways to pick the value of } \mu_{i,k}(n+1)}.$$

For the first term notice that there are n ways to choose where the fixed point is in the permutation of length n and then there are $f(n-1)$ ways to permute the remaining $(n-1)$ numbers in such a way that there are no fixed points.

For the second term notice that the number of permutations of length n with no fixed points is exactly $f(n)$. So we derive the recurrence relation,

$$f(n+1) = nf(n-1) + nf(n).$$

We can compute by hand that $f(2) = 1, f(3) = 2$, then using our recurrence

$$f(4) = 3(1) + 3(2) = 9$$

$$f(5) = 4(2) + 4(9) = 44$$

$$f(6) = 5(9) + 5(44) = 265$$

$$f(7) = 6(44) + 6(265) = 1854$$

so our answer is 1854. ■

Problem 8

In this question, we're going to prove *Lagrange's four square theorem*, which states that any positive integer can be written as the sum of four squares.

(a) Show that, if x and y are integers that can be written as the sum of four squares, xy can be written as the sum of four squares.

(b) Suppose that p is a prime number greater than 2. Show that there must exist a and b with $0 \leq a, b \leq \frac{p-1}{2}$ such that $a^2 + b^2 + 1 = np$ for some integer n .

Explain why $1 \leq n < p$.

(c) Suppose that m is the smallest positive integer such that mp can be written as the sum of four squares. Then, for a_1, a_2, a_3 and a_4 integers, we can write

$$mp = a_1^2 + a_2^2 + a_3^2 + a_4^2.$$

Can you find integers b_1, b_2, b_3 and b_4 such that $a_i \equiv b_i \pmod{m}$ and $b_1^2 + b_2^2 + b_3^2 + b_4^2 = mr$ with $0 \leq r \leq m$?

How can you write m^2pr as the sum of four squares? What about pr ?

(d) Hence prove Lagrange's four square theorem.

Solution

(a) Let $x = a_1^2 + a_2^2 + a_3^2 + a_4^2$. Let $y = b_1^2 + b_2^2 + b_3^2 + b_4^2$. Then

$$\begin{aligned} xy &= a_1^2 b_1^2 + a_1^2 b_2^2 + a_1^2 b_3^2 + a_1^2 b_4^2 \\ &\quad + a_2^2 b_1^2 + a_2^2 b_2^2 + a_2^2 b_3^2 + a_2^2 b_4^2 \\ &\quad + a_3^2 b_1^2 + a_3^2 b_2^2 + a_3^2 b_3^2 + a_3^2 b_4^2 \\ &\quad + a_4^2 b_1^2 + a_4^2 b_2^2 + a_4^2 b_3^2 + a_4^2 b_4^2. \end{aligned}$$

Now it can be (painstakingly) checked that

$$\begin{aligned} xy &= (a_1 b_1 + a_2 b_2 + a_3 b_3 - a_4 b_4)^2 \\ &\quad + (a_1 b_3 + a_2 b_4 - a_3 b_1 + a_4 b_2)^2 \\ &\quad + (-a_1 b_2 + a_2 b_1 + a_3 b_4 + a_4 b_3)^2 \\ &\quad + (a_1 b_4 - a_2 b_3 + a_3 b_2 + a_4 b_1)^2 \end{aligned}$$

and so xy can be expressed as the sum of four squares. $\square$

(b) Since the problem talks about "some" integer n , it suffices to show that there exist a, b satisfying those inequalities and $a^2 + b^2 \equiv -1 \pmod{p}$. I am going to show that it suffices to solve the problem for $0 \leq a, b \leq p-1$. Observe that modulo p the following sets are the same

$$\{0, 1, \dots, p-1\} = \left\{0, \pm 1, \dots, \pm \frac{p-1}{2}\right\}.$$

Now let $0 \leq r \leq p-1$ be an integer. Due to our observation, modulo p we can write $r \equiv k$ for some $0 \leq |k| \leq \frac{p-1}{2}$. So $r^2 \equiv k^2 \equiv |k|^2 \pmod{p}$. Now if we pick $0 \leq a \leq p-1$ we have shown by our previous

argument that there must be some integer $0 \leq x \leq \frac{p-1}{2}$ such that $a^2 \equiv x^2 \pmod{p}$, and we can say the same exact statement again just replacing a with b . Thus it is okay to just consider $0 \leq a, b \leq p-1$.

Now we are going to handle two cases, when $p \equiv 1 \pmod{4}$ and when $p \equiv 3 \pmod{4}$, since p is an odd prime.

Case 1: $p \equiv 1 \pmod{4}$. Then for some integer k we may write $p = 4k + 1$. We are going to pick $a = 0$ and then show that $b^2 \equiv -1$ has a solution mod p .

Claim 1: -1 is a quadratic residue modulo p .

Proof: We are going to use the Legendre symbol and Euler's criterion.

$$\begin{aligned} \left(\frac{p-1}{p} \right) &\equiv (p-1)^{\frac{p-1}{2}} \pmod{p} \\ &\equiv (-1)^{\frac{4k}{2}} \equiv (-1)^{2k} \equiv 1 \pmod{p}. \end{aligned}$$

Since the Legendre symbol evaluated to 1 mod p and $-1 \equiv 1 \pmod{p}$ only when $p = 2$ and we have $p > 2$ we can conclude that $\left(\frac{p-1}{p} \right) = 1$ and so, by definition of the Legendre symbol, $p-1 \equiv -1$ is a quadratic residue mod p . $\square$

Now set $a = 0$. Now we just need to find $0 \leq b \leq p-1$ such that $b^2 \equiv -1 \pmod{p}$, but such a b is guaranteed to exist due to claim 1.

Case 2: $p \equiv 3 \pmod{4}$. We may write $p = 4k + 3$ for some integer k . We are trying to find a pair of quadratic residues of the form $(a^2, b^2) = (q, -1 - q) \pmod{p}$, and we are going to show that such a pair must exist via the Pigeonhole principle.

Claim 2: Amongst the non-zero residues mod p , $\{1, 2, \dots, p-1\}$ there are exactly $\frac{p-1}{2}$ quadratic residues mod p .

Proof: Let $1 \leq a \leq p-1$ be an integer. Using the Legendre symbol and Euler's criterion we find

$$\begin{aligned} \left(\frac{a}{p} \right) &\equiv a^{\frac{p-1}{2}} \equiv a^{2k+1} \pmod{p} \\ \left(\frac{p-a}{p} \right) &\equiv (-a)^{\frac{p-1}{2}} \equiv -a^{2k+1} \pmod{p} \end{aligned}$$

Now Euler's criterion guarantees that both expressions evaluate to one of $\{-1, 1\}$. Since the two expressions sum to 0 mod p and $p > 2$ we can conclude that exactly one of the two expressions evaluates to 1. Thus, by definition of the Legendre symbol, exactly one of $(a, -a)$ is a quadratic residue mod p . Now since the non-zero residues mod p can be expressed in the form $\{\pm k \mid k = 1, 2, \dots, \frac{p-1}{2}\}$ we know that for each k precisely one of $(k, -k)$ is a square mod p , and thus there are exactly $\frac{p-1}{2}$ quadratic residues amongst the non-zero residues mod p . $\square$

Now since 0 is a quadratic residue, we can say that amongst $\{0, 1, \dots, p-1\}$ there are exactly $\frac{p-1}{2} + 1 = \frac{p+1}{2}$ squares mod p due to claim 2, and consequently $\frac{p-1}{2}$ non-quadratic residues. Consider pairs of numbers $(k, -1 - k)$ where k is a square mod p . Suppose FTSOC that over all $\frac{p+1}{2}$ choices of k the number $-1 - k$ is a non-quadratic residue. Note that $-1 - k_1 \equiv -1 - k_2 \iff k_1 \equiv k_2 \pmod{p}$, so we have found $\frac{p+1}{2}$ pairwise distinct non-quadratic residues mod p , but there should only be $\frac{p-1}{2}$, a contradiction. Hence there has to

exist a pair of squares mod p of the form $(a^2, b^2) = (q, -1 - q)$.

Finally we explain the inequality on n . Since $a, b \leq \frac{p-1}{2}$ we have

$$\begin{aligned} a^2 + b^2 + 1 &\leq \left(\frac{p-1}{2}\right)^2 + \left(\frac{p-1}{2}\right)^2 + 1 \\ &= p^2 - 2p + 2 \\ np &\leq p^2 - 2p + 2 \\ n &\leq p - 2 + \frac{2}{p} \end{aligned}$$

and since $p \geq 3$ we have $n \leq p - \frac{4}{3} \Rightarrow n < p$. Also since $a, b \geq 0$, we have $0^2 + 0^2 + 1 \leq np$ and so $n \geq \frac{1}{p}$ but since $n \in \mathbb{Z}$ we have $n \geq 1$. $\square$

(c) Due to the division algorithm, we can write $a_i = d_i m + r_i$ for $(i = 1, 2, 3, 4)$, where $d_i \in \mathbb{Z}$ and integers $0 \leq r_i < m$. We have two cases.

Case 1: m is odd.

Then modulo m the following is true,

$$\{0, 1, \dots, m-1\} = \left\{0, \pm 1, \dots, \pm \frac{m-1}{2}\right\}.$$

Thus for each $0 \leq r_i < m$ there is some integer x_i satisfying $0 \leq |x_i| \leq \frac{m-1}{2}$ and $r_i \equiv x_i \pmod{m}$. Now for each $(i = 1, 2, 3, 4)$ define $b_i = x_i$. Then clearly $a_i \equiv b_i \pmod{m}$ and

$$\begin{aligned} b_1^2 + b_2^2 + b_3^2 + b_4^2 &\leq \frac{(m-1)^2}{4} + \frac{(m-1)^2}{4} + \frac{(m-1)^2}{4} + \frac{(m-1)^2}{4} \\ &= (m-1)^2 \\ &< m^2. \end{aligned}$$

Case 2: m is even.

Then modulo m the following is true,

$$\{0, 1, \dots, m-1\} = \left\{0, \pm 1, \dots, \pm \frac{m}{2}\right\}.$$

Thus for each $0 \leq r_i < m$ there is some integer x_i satisfying $0 \leq |x_i| \leq \frac{m}{2}$ and $r_i \equiv x_i \pmod{m}$. Now for each $(i = 1, 2, 3, 4)$ define $b_i = x_i$. Clearly $a_i \equiv b_i \pmod{m}$ and

$$\begin{aligned} b_1^2 + b_2^2 + b_3^2 + b_4^2 &\leq \frac{m^2}{4} + \frac{m^2}{4} + \frac{m^2}{4} + \frac{m^2}{4} \\ &= m^2. \end{aligned}$$

Now since $a_i \equiv b_i \pmod{m}$ and $a_1^2 + \dots + a_4^2 \equiv 0 \pmod{m}$, we find that $b_1^2 + \dots + b_4^2$ is divisible by m . Recall that from both cases we showed that $b_1^2 + \dots + b_4^2 \leq m^2$. Now for some integer r we may write

$$0 \leq b_1^2 + b_2^2 + b_3^2 + b_4^2 = mr \leq m^2$$

which implies that $0 \leq r \leq m$, as desired.

Noting that $m^2 pr = mp \times mr$, we can use the identity in (a) to write

$$\begin{aligned} m^2 pr &= (a_1 b_1 + a_2 b_2 + a_3 b_3 - a_4 b_4)^2 \\ &\quad + (a_1 b_3 + a_2 b_4 - a_3 b_1 + a_4 b_2)^2 \\ &\quad + (-a_1 b_2 + a_2 b_1 + a_3 b_4 + a_4 b_3)^2 \\ &\quad + (a_1 b_4 - a_2 b_3 + a_3 b_2 + a_4 b_1)^2. \end{aligned}$$

Now consider $mr = b_1^2 + \dots + b_4^2$. Note that if we replace $b_i \rightarrow -b_i$ the quantity mr remains unchanged. So we can choose the signs of the b_i 's as we wish. Let us replace b_4 with $-b_4$ and now notice that

$$\begin{aligned} m^2 pr &= (a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4)^2 \\ &\quad + (a_1 b_3 - a_2 b_4 - a_3 b_1 + a_4 b_2)^2 \\ &\quad + (-a_1 b_2 + a_2 b_1 - a_3 b_4 + a_4 b_3)^2 \\ &\quad + (-a_1 b_4 - a_2 b_3 + a_3 b_2 + a_4 b_1)^2. \end{aligned}$$

and then since $a_i \equiv b_i \pmod{m}$,

$$m^2 pr \equiv (mp)^2 + (0)^2 + (0)^2 + (0)^2 \pmod{m}.$$

and so every bracket is divisible by m . Thus for some integers c_1, c_2, c_3, c_4 it is possible to write

$$\begin{aligned} m^2 pr &= m^2 c_1^2 + m^2 c_2^2 + m^2 c_3^2 + m^2 c_4^2 \\ pr &= c_1^2 + c_2^2 + c_3^2 + c_4^2 \end{aligned}$$

and so we can also write pr as the sum of four squares. $\square$

(d) We begin with an important claim. Let the a_i 's and the b_i 's be defined as they were in (c).

Claim 1: Let p be a prime. It is possible to express p as the sum of four squares.

Proof: Note that $2 = 1^2 + 1^2 + 0^2 + 0^2$, so from here on assume that $p > 2$. Let m be the smallest positive integer such that mp can be expressed as the sum of four squares. Let us suppose FTSOC that $m > 1$. Owing to part (c) it is also possible to find an integer $0 \leq r \leq m$ such that pr can be expressed as the sum of four squares.

Case 1: $r = m$. Again looking at part (c), this happens only when m is even and $a_i \equiv \pm \frac{m}{2} \equiv \frac{m}{2} \pmod{m}$ (note that for even m we have $\pm \frac{m}{2} \equiv \frac{m}{2} \pmod{m}$) for every $(i = 1, 2, 3, 4)$. That means there must be a sequence of integers $(k_i)_{i=1}^4$ such that $a_i = k_i m + \frac{m}{2}$. Hence,

$$\begin{aligned} mp &= \left(k_1 m + \frac{m}{2}\right)^2 + \left(k_2 m + \frac{m}{2}\right)^2 + \left(k_3 m + \frac{m}{2}\right)^2 + \left(k_4 m + \frac{m}{2}\right)^2 \\ &= m^2(k_1^2 + k_2^2 + k_3^2 + k_4^2) + m^2(k_1 + k_2 + k_3 + k_4) + m^2 \end{aligned}$$

But this implies that $m^2 \mid mp \Rightarrow m \mid p$. Thus $m \in \{1, p\}$. Since we assumed that $m > 1$, the only possibility is that $m = p$. But now recall that part (b) guaranteed the existence of an integer $1 \leq n < p$ satisfying $np = a^2 + b^2 + 1^2 + 0^2$ for some $a, b \in \mathbb{Z}$, and so $m = p$ cannot be the smallest positive integer such that mp

can be written as the sum of four squares, a contradiction.

Case 2: $r = 0$. We have $b_1^2 + \dots + b_4^2 = 0$. If this happens then it must be the case that $b_1 = b_2 = b_3 = b_4 = 0$, because if any one of the b_i 's were > 0 then the sum of their squares would be > 0 also. Thus $a_i \equiv b_i \equiv 0 \pmod{m}$. Hence, using the fact that $m \mid a_i \Rightarrow m^2 \mid a_i^2$

$$m^2 \mid (a_1^2 + a_2^2 + a_3^2 + a_4^2) = mp \Rightarrow m \mid p.$$

Thus $m \in \{1, p\}$, and we can show that both cases yield a contradiction using the same argument found at the end of case 1.

Case 3: $0 < r < m$. Then in this case we have found a smaller positive integer r such that pr can be expressed as the sum of four squares, directly contradicting the definition of m as the smallest positive integer such that mp can be written as the sum of four squares.

In each case we found a contradiction, so it must be the case that $m = 1$ or equivalently p can be expressed as the sum of four squares. $\square$

Now via the Fundamental theorem of arithmetic, we know that every positive integer can be expressed as a product of some finite number of primes. Then by repeatedly applying the lemma shown in part (a) of this problem we can show that every positive integer can indeed be expressed as the sum of four squares. $\blacksquare$